PROJECT REPORT

Project title: Luxury Housing Sales Analysis using Python, SQL & Power BI

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Problem Statement

The luxury real estate market in Bangalore is highly competitive.
Developers and investors need insights into:

- Which locations generate higher revenue
- Sales trends across quarters
- Buyer behaviour (NRI vs Local)
- Impact of amenities, connectivity, and infrastructure on pricing

This project analyses luxury housing sales data to support data-driven decision making.

Dataset Description

- **Source**: Provided dataset (Luxury_Housing_Bangalore.csv)
- **Records**: Luxury housing transactions in Bangalore
- **Key Columns**:
    - Micro_Market
    - Project_Name
    - Developer_Name
    - Ticket_Price_Cr
    - Price_per_Sqft
    - Quarter_Number
    - Buyer_Type
    - NRI_Buyer
    - Amenity_Score
    - Connectivity_Score

Tools & Technologies Used

- **Python** → Data Cleaning & EDA
- **Libraries** → Pandas, NumPy, Matplotlib, Seaborn
- **SQL (MySQL)** → Data storage & querying
- **Power BI** → Data visualization & dashboard creation

Data Cleaning (Python)

Handled missing and null values

```
pip install pandas numpy sqlalchemy pymysql psycopg2-binary


import pandas as pd

import numpy as np

df = pd.read_csv("Luxury_Housing_Bangalore.csv")

df.head()

df.info()

df.shape

df.info()

df.describe()

df['Transaction_Type'].value_counts()

df['Micro_Market'].value_counts().head(10)
```

Removed duplicates

```
df['Unit_Size_Sqft'] = df['Unit_Size_Sqft'].fillna(df['Unit_Size_Sqft'].median())


# Remove 'Cr' and commas

df['Ticket_Price_Cr'] = (
```

```
    df['Ticket_Price_Cr']
    .astype(str)
    .str.replace('Cr', '', regex=False)
    .str.replace(',', '', regex=False)
)
df['Ticket_Price_Cr'] = pd.to_numeric(df['Ticket_Price_Cr'], errors='coerce')
# Fill missing price using median by Configuration
df['Ticket_Price_Cr'] = df.groupby('Configuration')['Ticket_Price_Cr']\
    .transform(lambda x: x.fillna(x.median()))

df['Amenity_Score'] = df['Amenity_Score'].fillna(df['Amenity_Score'].median())
df['Buyer_Comments'] = df['Buyer_Comments'].fillna("No Comments")
df.isnull().sum()

df.shape
df.info()
df.describe()
```

Exploratory Data Analysis (EDA)

#Average Price by Configuration#

```
df.groupby('Configuration')['Ticket_Price_Cr'].mean().sort_values(ascending=False)
```

#Builder Performance Snapshot#

```
df.groupby('Developer_Name')['Ticket_Price_Cr']\
  .agg(['count', 'mean', 'sum'])\
  .sort_values('sum', ascending=False)\
  .head(10)
```

#Correlation Check (Amenity vs Price)#

```
df['Ticket_Price_Cr'].quantile([0.01, 0.99])
```

#Feature Engineering#

 # Price per Sqft#

```
df['Price_per_Sqft'] = (df['Ticket_Price_Cr'] * 1e7) / df['Unit_Size_Sqft']
```

 # Quarter Number#

```
quarter_map = {'Q1':1, 'Q2':2, 'Q3':3, 'Q4':4}
df['Quarter_Number'] = df['Purchase_Quarter'].map(quarter_map)
```

 # Booking Flag#

```
df['Booking_Flag'] = df['Transaction_Type'].apply(lambda x: 1 if x == 'Booking' else 0)
```

Exported cleaned dataset

```
df.to_csv("luxury_housing_cleaned.csv", index=False)
```

SQL Implementation

google.colab

```
from google.colab import files
uploaded = files.upload()

import pandas as pd
df = pd.read_csv("luxury_housing_cleaned.csv")
print("File loaded successfully!")
df.head()

import sqlite3
# Create database file
conn = sqlite3.connect("luxury_housing.db")

# Create table SQL
```

```
create_table_sql = """
CREATE TABLE IF NOT EXISTS housing_sales (
    Property_ID INTEGER,
    Micro_Market TEXT,
    Project_Name TEXT,
    Developer_Name TEXT,
    Unit_Size_Sqft REAL,
    Configuration TEXT,
    Ticket_Price_Cr REAL,
    Transaction_Type TEXT,
    Buyer_Type TEXT,
    Purchase_Quarter TEXT,
    Connectivity_Score REAL,
    Amenity_Score REAL,
    Possession_Status TEXT,
    Sales_Channel TEXT,
    NRI_Buyer INTEGER,
    Locality_Infra_Score REAL,
    Avg_Traffic_Time_Min REAL,
    Buyer_Comments TEXT,
    Price_per_Sqft REAL,
    Quarter_Number INTEGER,
    Booking_Flag INTEGER
);
"""
conn.execute(create_table_sql)
print("Table created successfully!")
```

```
df.to_sql("housing_sales", conn, if_exists="append", index=False)
print("Data inserted into SQL table!")

import pandas as pd

pd.read_sql("SELECT COUNT(*) AS total_rows FROM housing_sales", conn)

pd.read_sql("""
SELECT AVG(Ticket_Price_Cr) AS avg_ticket_price
FROM housing_sales
""", conn)

pd.read_sql("""
SELECT Project_Name, Ticket_Price_Cr
FROM housing_sales
ORDER BY Ticket_Price_Cr DESC
LIMIT 10
""", conn)

pd.read_sql("""
SELECT Micro_Market, COUNT(*) AS total_sales
FROM housing_sales
GROUP BY Micro_Market
ORDER BY total_sales DESC
""", conn)

final_df = pd.read_sql("SELECT * FROM housing_sales", conn)
final_df.to_csv("housing_sales_for_powerbi.csv", index=False)
```

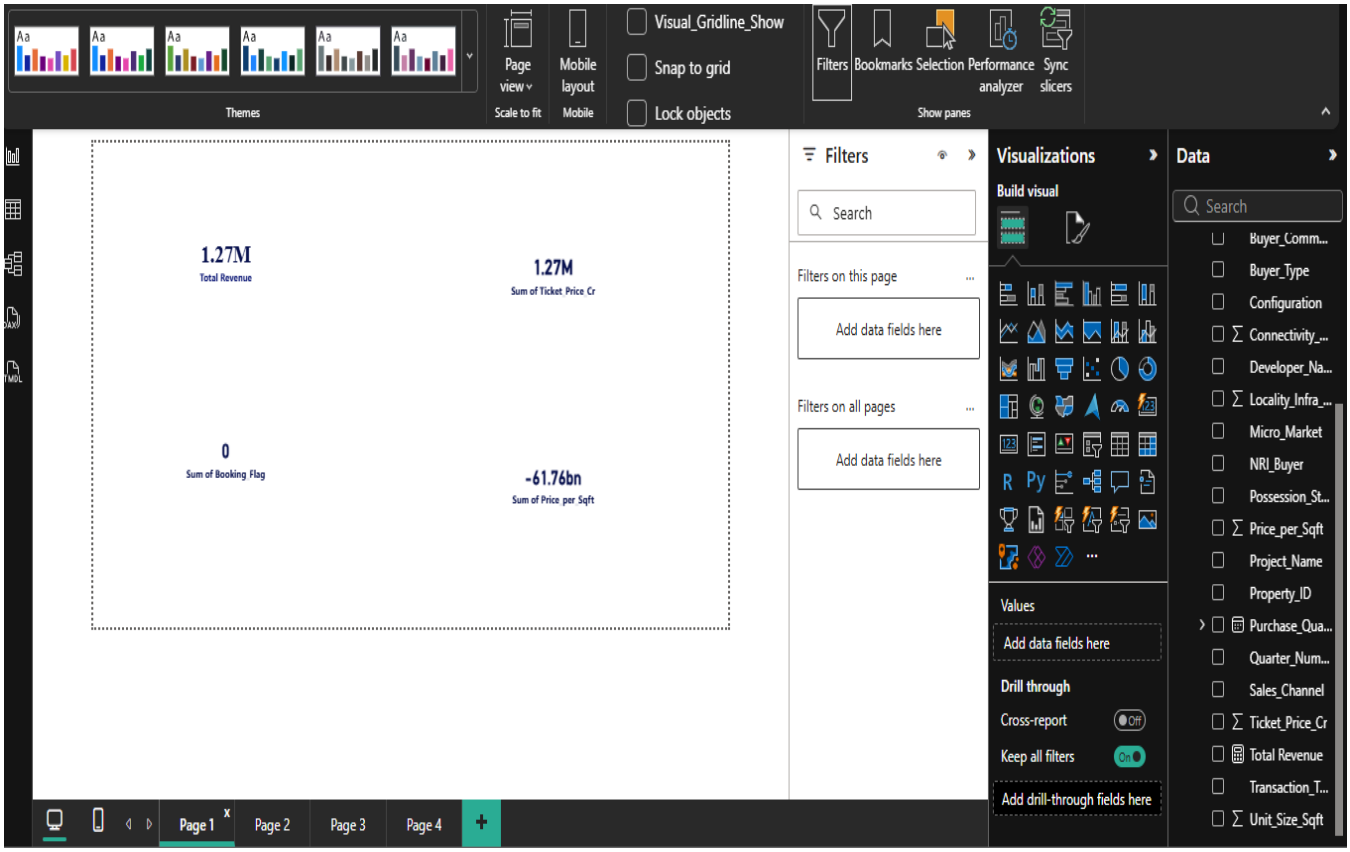
print("File ready for Power BI!")

Power BI Dashboard

Dashboard Pages:

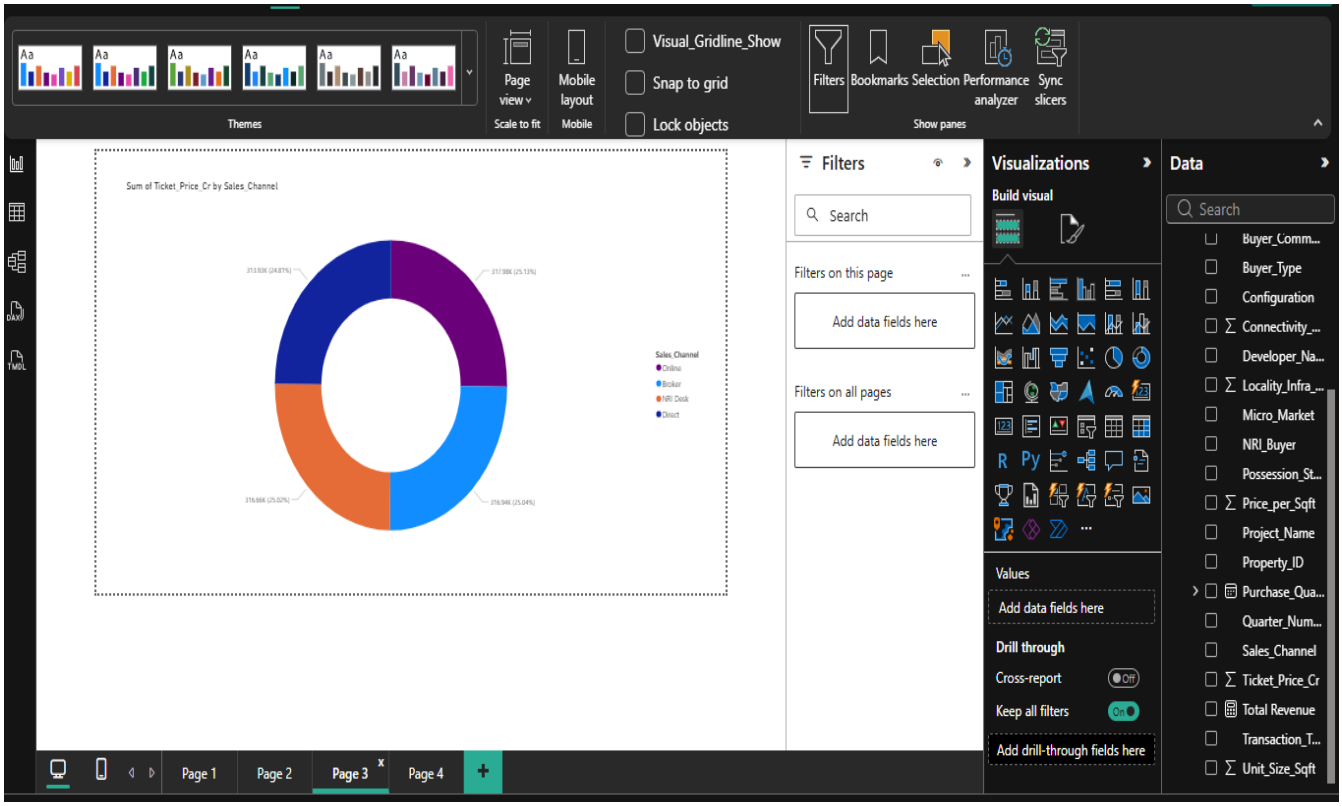
Page 1 – Overview

- KPI Cards:
    - Total Revenue
    - Average Ticket Price
    - Total Properties Sold

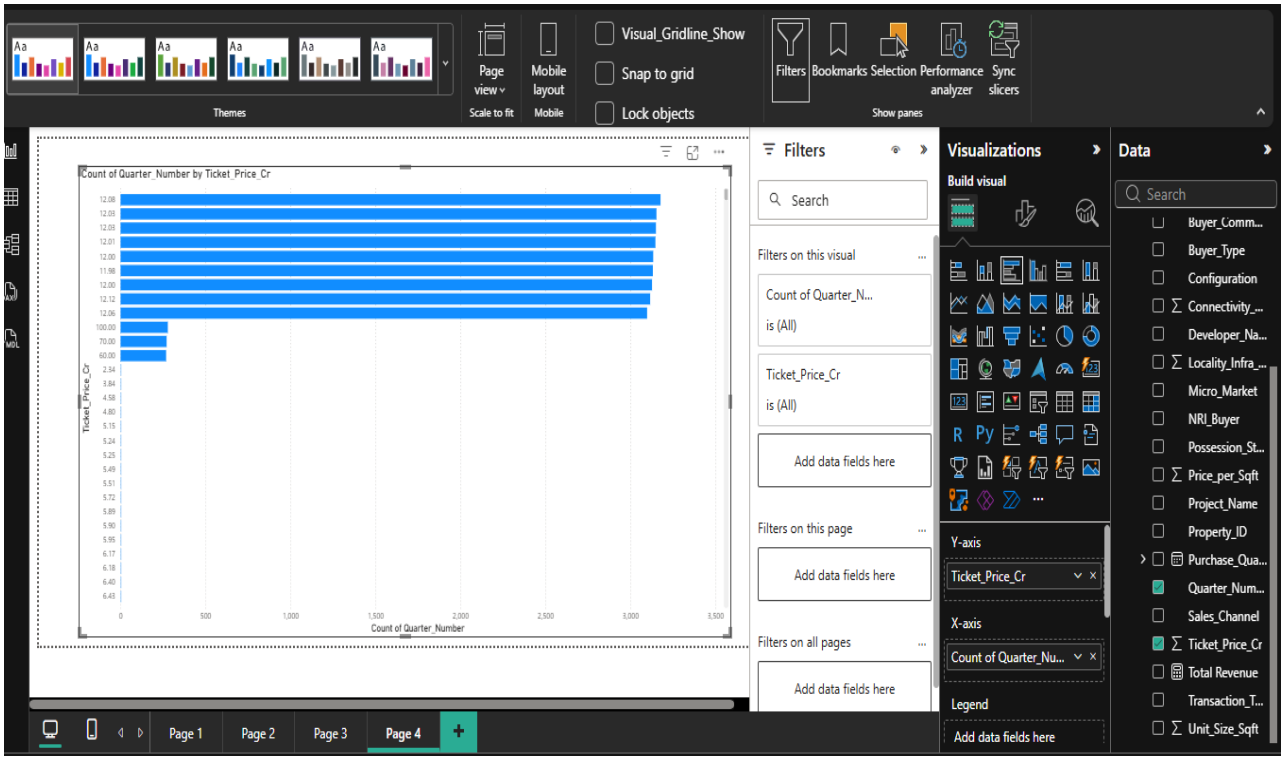


Page 2 – Donut chart

sales channels to total luxury housing revenue



Page 3 – Bar chart

Quarter-wise Transaction Distribution Across Ticket Price Ranges

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